Examiner only

2. The chemical formula of potassium carbonate is K_2CO_3.

(a) (i) State how many carbon atoms are present in the formula, K_2CO_3. .... [1]

(ii) Give the **total** number of atoms shown in the formula, K_2CO_3. .... [1]

(b) Calculate the percentage by mass of carbon in potassium carbonate, K_2CO_3. [2]

percentage by mass = ... %

4

Examiner only

2. Peter, Claire and David compared the hardness of three samples of water, **A**, **B** and **C**. They added soap solution to each sample until it gave a lather on shaking. They carried out a fair test.

(a) Give **two** ways they made it a fair test. [2]

....

....

....

(b) Their results are shown in the table.

Water sample	Volume of soap solution needed to give a lather (cm^3)			
	Peter	Claire	David	Mean
A	18	19	17	....
B	8	10	9	9
C	1	1	1	1

(i) Calculate the mean volume of soap solution needed for sample **A** to give a lather. **Show your working.** [2]

Mean = ... cm^3

(ii) Which of the three water samples is the **hardest**? Give a reason for your answer. [2]

....

....

....

Examiner only

(c) The box shows some advantages and some disadvantages of hard water.

strengthens bones and teeth	**forms limescale when heated**
wastes soap	**reduces heart disease**
forms a scum with soap	**tastes better**

Choose **all** the disadvantages of hard water from the box. [2]

....

....

....

....

(d) Choose from the box an ion that causes hardness in water. [1]

3430U201
05

K^+	**Na^+**	**Ca^{2+}**	**Cu^{2+}**

Ion ....

9

Examiner only

2. Diagrams **A**, **B**, **C** and **D** represent hydrogen (H_2), sulfur dioxide (SO_2), hypochlorous acid (HClO) and phosphine (PH_3) but **not in that order**.

A **B** **C** **D**

(a) (i) Give the **letter** of the diagram that represents an element. Give a reason for your answer. [2]

Letter ....

Reason ....

....

(ii) Give the **letter** of the diagram that represents SO_2. [1]

....

(iii) Use diagrams **A**, **B**, **C** and **D** above to work out which of the following diagrams represents water (H_2O). **Circle** the correct diagram. [1]

Examiner only

(b) (i) Calculate the relative molecular mass (M_r) of hypochlorous acid, HClO. [1]

A_r(H) = 1 A_r(O) = 16 A_r(Cl) = 35.5

M_r = ...

(ii) The relative molecular mass (M_r) of sulfur dioxide is 64.

Calculate the percentage by mass of sulfur in sulfur dioxide, SO_2. [2]

A_r(S) = 32

Percentage = ... %

(c) Phosphine (PH_3) is made when phosphorus and hydrogen react.

Put a tick (✓) in the box next to the correctly balanced symbol equation for this reaction. [1]

$2P + 3H_2 \longrightarrow PH_3$	☐
$2P + 3H_2 \longrightarrow 2PH_3$	☐
$2P + 3H_2 \longrightarrow 3PH_3$	☐
$2P + 3H_2 \longrightarrow 4PH_3$	☐

8

Examiner only

2. A year 10 class investigated the reactions between some metals and hydrochloric acid. Their results are summarised in the table below.

Metal	Initial temperature (°C)	Final temperature (°C)	Rise in temperature (°C)	General observations
zinc	21	32	11	a few bubbles
calcium	22	66	44	lots of bubbles, solution spills out of test tube
magnesium	20		31	lots of bubbles
copper	21	21	0	no bubbles
iron	22	25	3	one or two bubbles

(a) (i) State which metal did **not** react with hydrochloric acid.

Give a reason for your choice. [1]

Metal ....

Reason ....

(ii) Calculate the final temperature for the reaction between magnesium and hydrochloric acid. [1]

Final temperature = ... °C

(b) What name is given to a reaction which gives a rise in temperature? [1]

Choose your answer from the box.

endothermic	**combustion**	**exothermic**	**precipitation**

....

3430U201
05

Examiner only

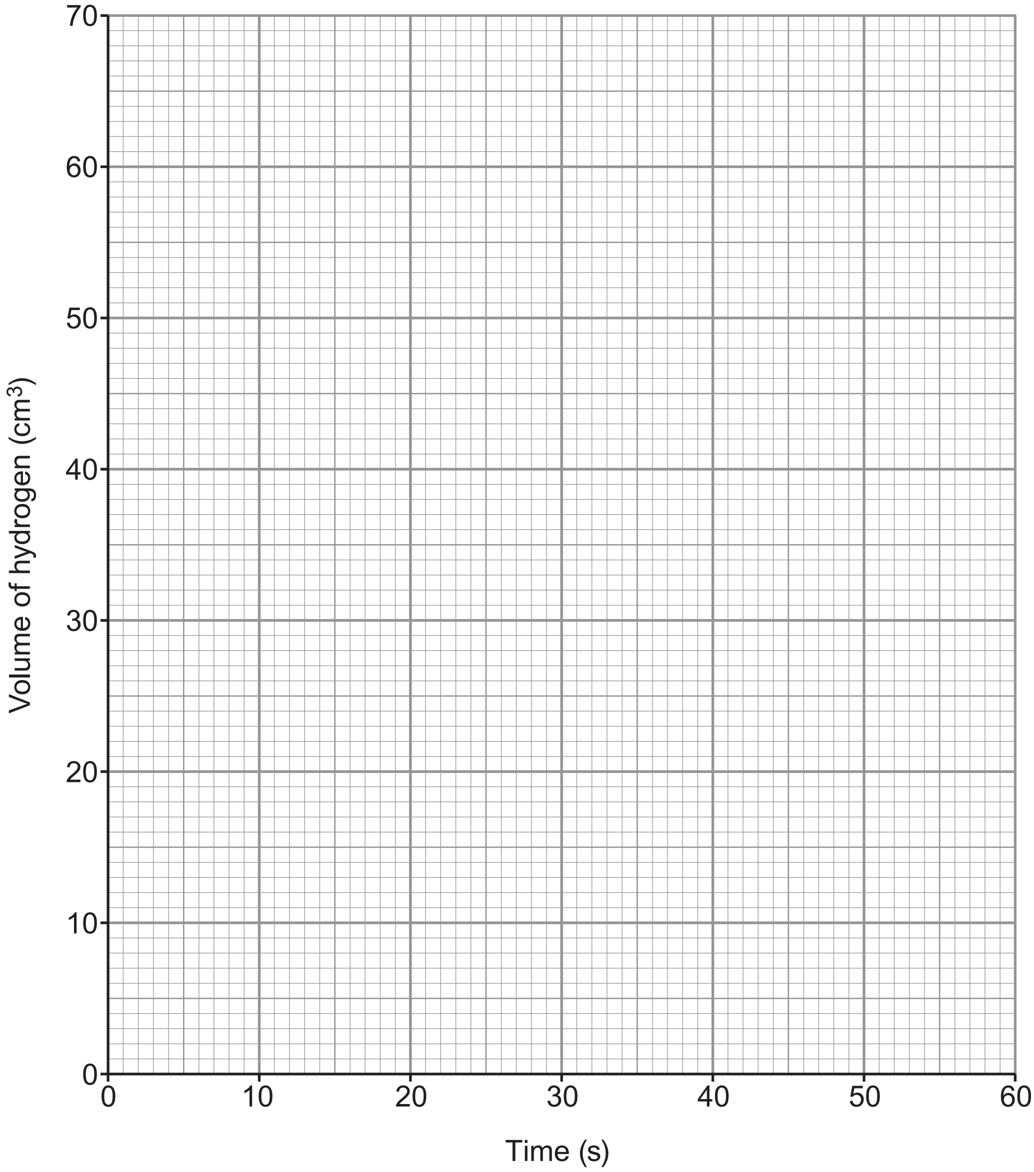
(c) The year 10 class then decided to investigate the rate of the reaction between magnesium and hydrochloric acid.

A piece of magnesium was placed in excess hydrochloric acid at 20 °C. The volume of hydrogen produced was recorded every 10 s.

The results obtained are shown in the table.

Time (s)	0	10	20	30	40	50	60
Volume of hydrogen (cm^3)	0	19	31	40	47	53	56

(i) Plot the volume of hydrogen against time on the grid. Draw a suitable line. [3]



Examiner only

(ii) The reaction had not finished after 60 s. How does the graph show this?

Put a tick (✓) in the correct box. [1]

Graph stops at 60 s	☐
Graph is still rising at 60 s	☐
Graph reaches a maximum temperature of 56 °C	☐

(iii) Why does the reaction slow down over time?

Put a tick (✓) in the correct box. [1]

The particles collide with less energy so less chance of successful collisions	☐
The particles move slower so less chance of successful collisions	☐
The particles have less surface area so less chance of successful collisions	☐
The particles get used up so less chance of successful collisions	☐

(iv) Suggest **two** changes you could make to the hydrochloric acid to make the reaction faster. [2]

....

....

(v) In the reaction between magnesium and hydrochloric acid, magnesium chloride is formed. Magnesium chloride contains Mg^{2+} ions and Cl^{-} ions.

Give the formula of magnesium chloride. [1]

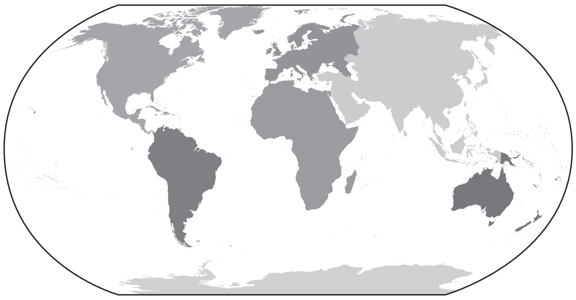
....

3430U201
07

11

Examiner only

2. The diagram shows how the Earth's continents are arranged today.



In 1912, Alfred Wegener proposed the theory of continental drift.

(a) (i) **Use statements from the table to answer parts I. and II. below.**

A	there are similar animals on different continents
B	there are similar fossils on different continents
C	only Europe, Asia and Africa were once joined
D	all of the continents were once joined
E	there is a jigsaw fit of different continents
F	there are similar rocks on different continents
G	none of the continents were joined

I. Give the **letter** of the statement that describes how Wegener suggested the continents were originally arranged. [1]

....

II. Give the **letters** of **three** pieces of evidence that Wegener used to support his theory. [3]

....

(ii) Underline the correct word in the brackets to complete the following sentence. [1]

Wegener could not explain how the continents moved but it is now known that this happens because of **(conduction / convection / radiation)** currents in the Earth's mantle.